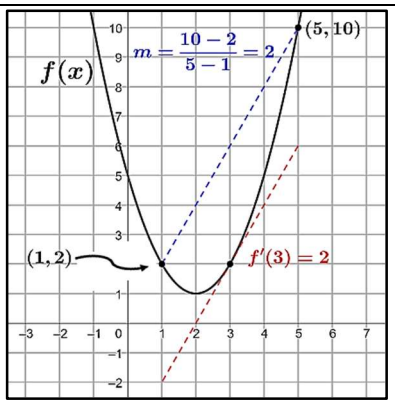
FUN-1.B.1 If a function f is continuous over the interval [a, b] and differentiable over the interval (a, b), then the Mean Value Theorem guarantees a point within that open interval where the instantaneous rate of change equals the average rate of change over the interval.

5-1 The Mean Value Theorem

Get Ready…

Answer the questions in the table below:

Let $f(x) = x^2 - 4x + 5$.	
Average Rate of Change (AROC) Given $f(x)$ over the interval $[a, b]$, the AROC of $f(x)$ is $\text{AROC} = \dfrac{f(b) - f(a)}{b - a}$	**Instantaneous Rate of Change (IROC)** Given $f(x)$ over the interval $[a, b]$, the IROC of $f(x)$ at $x = c$ is $\text{IROC} = f'(c)$
Find the average rate of change of $f(x)$ over the interval $[1, 5]$.	Find the instantaneous rate of change of $f(x)$.
Is there a value of x where the instantaneous rate of change of $f(x)$ equals the average rate of change of $f(x)$ over the interval $[1, 5]$?	

In the example above, we were able to find a value in the given interval where AROC = IROC. It is natural to wonder if this is always the case-can we always find a place where $f'(x)$ equals the average rate of change of $f(x)$ in a given interval?

This is the basis for the Mean Value Theorem.

The Mean Value Theorem (MVT)	If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then there exists a number c in (a, b) such that $$f'(c) = \frac{f(b) - f(a)}{b - a}$$ (or IROC = AROC somewhere in the interval)

<u>Big Ideas about the Mean Value Theorem (MVT):</u>

1. The function f must be continuous AND differentiable on the given interval (closed-open).
2. MVT guarantees that IROC = AROC somewhere in the interval.
3. MVT is an **<u>existence theorem</u>**…it guarantees something will occur but does not specify **<u>where</u>** it occurs.
4. We will approach MVT problems in a similar way as we did IVT:
   check conditions, use endpoints, state conclusion.

Let's compare the MVT and the IVT.

<table>
<tr><th>MVT</th><th>IVT</th></tr>
<tr><td>Conditions:
1.) f is continuous
2.) f is differentiable</td><td>Conditions:
1.) f is continuous</td></tr>
<tr><td>What to look for:
Conclusion about the <u>rate of change of the function</u></td><td>What to look for:
Conclusion about the <u>values of the function</u></td></tr>
<tr><td>Example:
<table>
<tr><td>t (min)</td><td>0</td><td>1</td><td>5</td><td>9</td></tr>
<tr><td>$v(t)$ (ft/min)</td><td>5</td><td>2</td><td>1</td><td>-4</td></tr>
</table>
A bug is moving along the x-axis. The bug's velocity is differentiable, and selected values are given in the table above. Is there a time c, for $0 < c < 9$, such that $a(c) = -1$? Give a reason for your answer.</td><td>Example:
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A bug is moving along the x-axis. The bug's velocity is differentiable, and selected values are given in the table above. Is there a time c, for $0 < c < 9$, such that $\mathrm{v}(c) = -1$? Give a reason for your answer.</td></tr>
<tr><td>*Work:
v is differentiable on $(0, 9)$
So v is continuous on $[0, 9]$
$a(c) = v'(c)$
$\frac{v(9) - v(0)}{9 - 0} = \frac{-4 - 5}{9} = \frac{-9}{9} = -1$</td><td>*Work:
v is differentiable on $(0, 9)$
So v is continuous on $[0, 9]$
$v(c) = -1$ is between $v(9) = -4$ and $v(0) = 5$</td></tr>
<tr><td>*Statement:
Since _____ is _____ on _____ and _____ on _____, the _____ guarantees that…</td><td>*Statement:
Since _____ is _____ on _____ the _____ guarantees that…</td></tr>
</table>

*The work alone or statement alone are **NOT** sufficient enough to receive credit for the problem—you must have BOTH!!!!!!!

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x	0	2	3	6
$g(x)$	-2	3	1	10

<u>Example 1:</u> The function g is differentiable for all values of x. Selected values of $g(x)$ are given in the table above. Is there a time c, $0 < c < 6$, such that $g'(c) = 2$? Give a reason for your answer.

x	2	6	9	15
$f'(x)$	1	8	-4	6

Example 2: The function f' is differentiable for all values of x. Selected values of $f'(x)$ are given in the table above. Can you guarantee a time k, $2 < k < 15$, such that $f'(k) = 5$? Give a reason for your answer.

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t minutes	0	5	15	20	30
$h(t)$ feet	0	40	70	65	80

Example 3: A hot air balloon is launched into the air with a human pilot. The twice-differentiable function h models the balloon's height, measured in feet, at time t, measured in minutes. The table above gives values of the height of the balloon at selected times t.

(a) For $5 \leq t \leq 15$, must there be a time t when the balloon is 50 feet in the air? Justify your answer.

(b) For $20 \leq t \leq 30$, must there be a time t when the balloon's velocity is 1.5 feet per minute? Justify your answer.